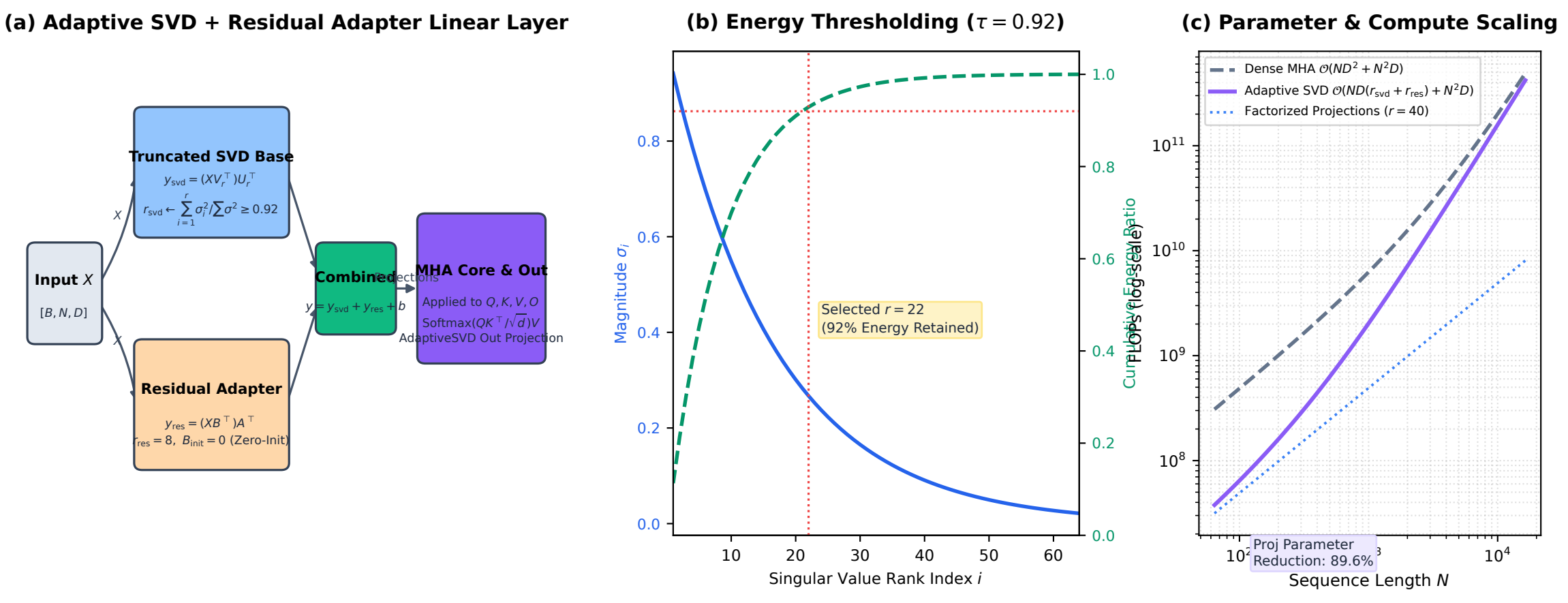
(a) Adaptive SVD + Residual Adapter Linear Layer
Input X
$[B, N, D]$
X
X
Truncated SVD Base
$y_{\text{svd}} = (XV_r^\top)U_r^\top$
$r_{\text{svd}} \leftarrow \sum_{i=1}^{r} \sigma_i^2 / \sum \sigma^2 \geq 0.92$
Residual Adapter
$y_{\text{res}} = (XB^\top)A^\top$
$r_{\text{res}} = 8,\ B_{\text{init}} = 0$ (Zero-Init)
Combined Projections
$y = y_{\text{svd}} + y_{\text{res}} + b$
MHA Core & Out
Applied to Q, K, V, O
$\text{Softmax}(QK^\top/\sqrt{d})V$
AdaptiveSVD Out Projection
(b) Energy Thresholding ($\tau = 0.92$)
Magnitude σ_i
Cumulative Energy Ratio
Selected $r = 22$
(92% Energy Retained)
Singular Value Rank Index i
(c) Parameter & Compute Scaling
Dense MHA $\mathcal{O}(ND^2 + N^2D)$
Adaptive SVD $\mathcal{O}(ND(r_{\text{svd}} + r_{\text{res}}) + N^2D)$
Factorized Projections ($r = 40$)
FLOPs (log-scale)
Proj Parameter
Reduction: 89.6%
Sequence Length N